Session 5: Who are the star products?

Question: Which products bring most of the money?

Concepts

Concept	In daily life	In Excel	The limit
groupby + sum	Sorting receipts into piles, then adding each pile	PivotTable	reset_index() makes it a normal table again
cumsum	A running total on a receipt roll	=SUM(B2:B2) dragged down	Sort first, or it means nothing
ABC analysis	A few items take most of the money	Sort, running %, IF	80 and 95 are a habit, not a law
.apply(function)	Using your rule on every row	Dragging your formula down	Use a named function

Code patterns

```
by_product = sales_clean.groupby("stock_code")["revenue"].sum()
by_product = by_product.sort_values(ascending=False)
product_value = by_product.reset_index()

product_value["cumulative_share"] = product_value["revenue"].cumsum() / total_revenue

product_value["abc_class"] = product_value["cumulative_share"].apply(abc_class)
(product_value["abc_class"] == "A").sum()
```

Three common errors

Error	What to do
Index looks strange	You forgot reset_index(). Add it after the groupby.
Running total makes no sense	You used cumsum before sorting. Sort first.
apply with a lambda	We use a named function in this course. Write def first.

Exercises and expected answers

Exercise	Expected
Predict: How many products are there? One row per stock code. Guess first: more or fewer than 5,000?	Guess first, then run it to check
Fill in: Revenue per country, biggest first.	United Kingdom on top with 1591694.5
Fix it: Germany's revenue. One name is wrong since the rename in Session 4.	37365.63
Fill in: A running total of revenue, in a new column.	the second row is 82285.83
Predict: A product sits at a cumulative share of 0.90. Which class? Guess first.	Guess first, then run it to check
Write it: Count the class B products and the class C products.	794 and 1601
Write it: The share of products in class A, in percent with 1 decimal.	21.9
Write it: The share of revenue from class A, in percent with 1 decimal. Then one f-string sentence.	21.9% of products bring 80.0% of revenue
Extra: Revenue for each month, for all countries.	775572.23, 593178.39, 507783.42
Extra: The top 10 products' share of revenue, in percent with 2 decimals.	9.62

Cold test scoring (out of 4, one point each)

Point	What
1	The code runs
2	The numbers are right
3	The names are clear
4	One sentence says what it means